

ABES ENGINEERING COLLEGE, GHAZIABAD

**(An Autonomous Institution Affiliated to Dr. A.P.J. Abdul Kalam Technical
University, Lucknow, Uttar Pradesh)**



EVALUATION SCHEME

For

B. Tech. 2ND Year

Computer Science & Engineering

(Effective for the students admitted into 1ST year from 2025-26 onwards)

Academic Year 2026-27



ABES Engineering College, Ghaziabad
(An Autonomous Institute)
B. Tech. (CSE) 2nd Year Evaluation Scheme

SEMESTER-III											
Bucket	Subject Code	Subject	Periods			Internal Evaluation		External Evaluation		Total	Credits
			L	T	P	Theory	Practical	Theory	Practical		
Mathematics and Problem- Solving	25AS301/ 25OE301----- 25OE310	Applied Maths for Computing Applications/ Science Based Open Elective (SBE)	2	1	0	30		70		100	3
Academic Curriculum Core	25CS301	Object Oriented Programming Paradigm	3	0	0	30		70		100	3
Academic Curriculum Core	25CS302	Operating System	3	0	0	30		70		100	3
Academic Curriculum Core	25CS303	Advanced Data Structure	3	0	0	30		70		100	3
Professional Elective	25VA301/ 25VA302	Data Visualization using Python/ Fundamentals of Cloud Computing	2	0	2	30		70		100	3
Ethical Legal and Social Aspects (Multidisciplinary / Humanities)	25HU301/ 25HU302	Universal Human Values/Technical Communication	2	0	0	30		70		100	2
Academic Curriculum Core	25CS351	Object Oriented Programming Paradigm Lab	0	0	2		50		50	100	1
Academic Curriculum Core	25CS352	Operating System Lab	0	0	2		50		50	100	1
Academic Curriculum Core	25CS353	Advanced Data Structure Lab	0	0	2		50		50	100	1
Basic Engineering Course	25VA351	FSD Workshop--I	0	0	2		50		50	100	1
Skill - centric Courses	25VA309	Employability Skills (logical Reasoning , Quant, Apti etc)	0	0	2		50		50	100	1
Innovation, Entrepreneurship, Industry Connect for Employability Skills	25VA352	Holistic Skill & Innovation III	0	0	2		100			100	1
Total										1200	23

Emerging Technology Elective:

- | | |
|---|--|
| 1. Track – AI: | 25VA301: Data Visualization using Python |
| 2. Track – Cloud: | 25VA302: Fundamentals of Cloud Computing |
| 3. Track – VLSI: | 25VA303: Electronic Circuits using SPICE Modelling |
| 4. Track – EV: | 25VA304: Basic of Electric Mobility & Vehicle Architecture |
| 5. Track – Industrial Automation: | 25VA305: Industrial Automation using PLC |
| 6. Track – Digital Automotive Technology: | 25VA306: Fundamentals of Digital Automotive & Design |



ABES Engineering College, Ghaziabad
(An Autonomous Institute)
B. Tech. (CSE) 2nd Year Evaluation Scheme

SEMESTER-IV											
Bucket	Subject Code	Subject	Periods			Internal Evaluation		External Evaluation		Total	Credits
			L	T	P	Theory	Practical	Theory	Practical		
Mathematics and Problem- Solving	25AS401/ 25OE401----- 25OE410	Applied Maths for Computing Applications/ Science Based Open Elective (SBE)	2	1	0	30		70		100	3
Academic Curriculum Core	25CS401	Computer Architecture & Organization	3	0	0	30		70		100	3
Academic Curriculum Core	25CS402	Database Management system	3	0	0	30		70		100	3
Academic Curriculum Core	25CS403	Design & Analysis of Algorithm	3	0	0	30		70		100	3
Professional Elective	25VA401/ 25VA402	Data Handling and Artificial Intelligence / Design and Development of Cloud Services	2	0	2	30		70		100	3
Ethical Legal and Social Aspects (Humanities / Multidisciplinary)	25HU401/ 25HU402	Universal Human Values/Technical Communication	2	0	0	30		70		100	2
Academic Curriculum Core	25CS451	Computer Organization & Architecture Lab	0	0	2		50		50	100	1
Academic Curriculum Core	25CS452	Database Management System Lab	0	0	2		50		50	100	1
Academic Curriculum Core	25CS453	Design & Analysis of Algorithm Lab	0	0	2		50		50	100	1
Skill centric/Innovation, Entrepreneurship and Industry Connect (Multidisciplinary)	25VA451	FSD Workshop--II	0	0	2		50		50	100	1
Skill - centric Courses	25VA409	Employability Skills II	0	0	2		50		50	100	1
Innovation, Entrepreneurship and Industry Connect	25VA452	Holistic skill & Innovation IV	0	0	2		100			100	1
Total										1200	23

Emerging Technology Elective:

- | | | |
|----|--|---|
| 1. | Track – AI: | 25VA401: Data Handling and Artificial Intelligence |
| 2. | Track – Cloud: | 25VA402: Design and Development of Cloud Services |
| 3. | Track – VLSI: | 25VA403: VLSI System Design using Verilog and FPGA |
| 4. | Track – EV: | 25VA404: Vehicle Dynamics and EV Power Requirements |
| 5. | Track – Industrial Automation: | 25VA405: Industrial Automation based on Pneumatics |
| 6. | Track – Digital Automotive Technology: | 25VA406: Automotive Product Design and Development |

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	III	Object Oriented Programming System	25CS301	3	0	0	3
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
36	0	0	54		90		

Course Objective (s)

The course aims:

1. Build strong understanding of OOP principles
2. Apply OOP concepts to real-world problems.
3. Develop modular, reusable and maintainable code.
4. Bridge theory with modern programming practices.
5. Implement generic programming in problem solving.

Syllabus

Unit No.	Topics	CI Hours
1	Evolution of Programming paradigms (monolithic, procedural, structured, OOP), Structured vs Object Oriented Development, Structure of C++ program, basics of classes and objects, scope resolution operator (::), reference variable, function, parameter passing (value, reference and address) , inline functions, function overloading, default arguments, function with variable number of arguments (variadic functions), auto keyword, range based for loop	8
2	Accessing class members, defining member functions, outside member functions as inline, data hiding, objects as arguments and as return from function, friend keyword: functions and classes, constant parameters, static keyword: data and member functions, constructors and its type, destructor, constant object and constructor, static data members with constructors and destructors, nested classes, pointers to objects and dynamic object management, array of objects , this pointer, self-referential classes, unique_ptr, shared_ptr, nullptr	8
3	Concept of polymorphism, types of polymorphism : (compile-time and runtime), function overloading, function overriding, constructor overloading and chaining, Overloadable operators, unary and binary operator overloading, operator	8

	overloading using member and friend functions, data conversion between basic to object and vice versa	
4	Concepts of Inheritance, forms of inheritance (single, multilevel, multiple, hierarchical), benefits and cost of inheritance, Composition vs inheritance, inheritance and member accessibility, constructors and destructors in derived classes, constructors invocation and data member initialization, function overriding, virtual functions and runtime polymorphism, base class pointer and derived object behavior, pure virtual function, abstract class and virtual destructors, rules for virtual functions.	6
5	Function templates and class templates, function overloading with templates, multiple parameters and generic programming concepts, Standard Template Library (STL): overview of containers, iterators and algorithms. Containers (pair, vector, set, map, stack, queue), iterators and their types, algorithms (sort, find, basic operations), function objects (comparator, lambda expressions).	6

Prescribed Textbooks					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Object Oriented Programming with C++	Reema Thareja	Revised 1 st Ed	Oxford University Press	2018
2	Mastering Object-Oriented Programming With C++	RS Salaria	6 th	Khanna Publishing	2017
3	Mastering C++	K R Venugopal, Rajkumar, T Ravishankar	25 th Reprint	McGraw Hill Education	2006
4	Object Oriented Programming with C++	E Balagurusamy	9 th	McGraw Hill	2026
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	C++ Primer	Stanley B. Lippman, Stanley Lippman, Barbara Moo	5	Addison-Wesley Professional	2013
2	The C++ Programming Language	Bjarne Stroustrup	4	Addison-Wesley	2013
3	Head First Object-Oriented Analysis & Design	Brett D. McLaughlin, David West, Gary Pollice	1	Shroff/O'Reilly	2011
Web References					
1. https://nptel.ac.in/courses/106105234					
2. https://nptel.ac.in/courses/106105151					

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs):		
Upon the completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Understand fundamental concepts of OOP and C++ constructs	K2
CO2	Implement the concepts of classes, objects and memory management techniques	K3
CO3	Implement concepts of polymorphism	K3
CO4	Apply the concepts of inheritance and virtual function	K3
CO5	Analyse C++ programs with STL and containers	K4

KL- Bloom's Knowledge Level (K_1 , K_2 , K_3 , K_4 , K_5 , K_6)

K_1 - Remember, K_2 - Understand, K_3 - Apply, K_4 - Analyze, K_5 - Evaluate, K_6 - Create

CO-PO & PSO Mapping:

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1		1			1		1				1			
CO2	1	1			1		1				1			
CO3		2	1		1		1				1			
CO4		2	1		1		1				1			
CO5	1	3	3		1		1	3	3		1			
Course Avg.	1.00	1.80	1.66		1		1	3	3		1.00			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz 1	100%	5
CO2	Quiz 2 + Assignment	50% + 50%	5
CO3	Quiz 3 + Class Test	40% + 60%	5
CO4	Quiz 4 + Class Test	40% + 60%	5
CO5	Quiz 5 + Certification Course	30% + 70%	5
	Attendance	100%	5

Total Marks		30
Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term – I	CO1 (10 marks) + CO2 (5 marks)	15
Mid Term -II	CO2 (5 marks) + CO3 (10 Marks)	15
End Term	CO1+CO2+CO3+CO4+CO5 (4+4+4+14+14 Marks)	40
Total Marks		70

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	III	Operating System	25CS302	3	0	0	3
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
36	0	0	54		90		

Course Objective (s)		
The course aims:		
1. To understand the fundamental concepts and architectures of operating systems.		
2. To analyze process management, scheduling, synchronization, and deadlock handling mechanisms.		
3. To apply memory management and virtual memory techniques for efficient resource utilization.		
4. To evaluate disk and file management techniques for effective storage management..		
Syllabus		
Unit No.	Topics	CI Hours
1	Unit: Introduction and Process Fundamentals Operating System Overview: Definition, Goals , Functions, Classification of Operating systems- Batch, Interactive, Time-sharing, Real-Time System. Multiprocessor Systems, Multiuser Systems, Multithreaded Systems, Multi-programming Vs Multi-tasking. Operating System Architectures: Kernel Mode, User Mode, System calls, Layered, Monolithic & Microkernel systems, Reentrant Kernel. Virtualization Basics: Virtual Machines, Containers, VM vs Container Comparison. Process Fundamentals: Job/Program/Process/Threads concepts, Process States Transition Diagram, Process Control Block (PCB) .	8
2	Unit: CPU Scheduling & Deadlock Management Scheduling Basics: Concept, Need ,CPU-I/O burst cycle, preemptive and non-preemptive scheduling, Context Switching, Schedulers, Dispatcher, Scheduling Criteria, Scheduling algorithms: FCFS, SJF, SRTF, Priority, Round Robin, Multilevel queue and Feedback scheduling, overview of Real-world scheduling systems (Linux/Android). Deadlock: System Model, Deadlock Characterization, Prevention, Avoidance(Banker's Algorithm) and Detection, Resource Allocation graph(RAG), Wait for graph(WFG), Recovery from deadlock, Starvation vs Deadlock.	8
3	Unit: Process Synchronization and Concurrency Concurrent Processes: Concurrent Processes, Principles of Concurrency, Race Condition, Critical Section Problem (CSP), Software Solution: Peterson's Algorithm, Hardware Solution: Test-and-Set, Mutex Locks, Semaphores (Binary & Counting), Inter-Process Communication (IPC): Shared Memory and Message Passing Models. Classical Synchronization Problems: Producer-Consumer Problem, Dining Philosophers Problem, Sleeping Barber Problem.	8

	Case Study: Web Servers Handling Concurrent Requests (Thread Pools, Request Queues).	
4	Unit: Memory Management and Virtual Memory Memory Management Basics: Logical and Physical Address Space, Address Binding, Dynamic Loading and Linking, Memory Protection. Contiguous Memory Allocation: Fixed and Variable Partitioning, Internal and External Fragmentation, Compaction, First Fit, Best Fit, Worst Fit. Non-Contiguous Memory Allocation: Paging, Page Table Structure, Address Translation, TLB, Segmentation, Segmentation with Paging. Virtual Memory: Virtual Memory Concept, Demand Paging, Page Fault Handling, Locality of Reference. Frame Allocation and Thrashing: Allocation Strategies, Global vs Local Allocation, Thrashing, Working Set Model. Case Study: Memory Usage in Modern Applications (Mobile Apps and Web Browsers).	6
5	Unit: Disk and File System Management Disk Management: Disk Structure, Disk Performance Parameters, Disk Scheduling Algorithms (FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK), Performance Analysis. RAID: RAID Concept and RAID Levels. File System: File Attributes (Metadata), File Operations, File Organization, Buffering and Caching, File Access Methods (Sequential and Direct Access). Directory Management: Directory Structure, Naming, Searching and Access Control. Reliability and Recovery: Backups, Journaling File Systems, Log-Structured File Systems.	6

Prescribed Textbooks					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Operating System Concepts	Silberschatz, Abraham, Peter B. Galvin, and Greg Gagne.	10th	John Wiley & Sons	2018
2	Modern Operating Systems	Andrew S. Tanenbaum	5th Edition	Pearson	2022
3	Operating Systems: Internals and Design Principles	William Stallings	10th Edition	Pearson	2018
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Modern Operating Systems	Andrew S. Tanenbaum	5th Edition	Pearson	2022
Web References					
1. NPTEL: Operating System Fundamentals :By Prof. Santanu Chattopadhyay https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs156 .					
2. UdemY Course: The Complete Operating Systems Course: From Zero to Hero! https://www.udemy.com/course/the-complete-operating-systems-course-from-zero-to-expert/?couponCode=UDEAFFHP22025 .					

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs):

Upon the completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Explain the fundamental concepts, architectures, and process management mechanisms of operating systems.	K2
CO2	Implement CPU scheduling algorithms and deadlock handling techniques for efficient process management.	K3
CO3	Apply synchronization techniques and inter-process communication mechanisms to solve concurrency problems.	K3
CO4	Apply memory management techniques, including paging, segmentation, and virtual memory, for efficient memory utilization.	K3
CO5	Apply disk scheduling techniques and file system structures for efficient storage management.	K3

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	2	1	3				1		2		3			
CO2	2	3	3	3	2	1			1		3			
CO3	2	3	3	3	2				1		3			
CO4	2	3	3	3	2		1		1		2			
CO5	2	3	3	3	2	1	1		2		2			
Course Avg.	2.00	2.60	3.00	2.40	1.60	0.40	0.60		1.40	0.00	2.60			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz 1	100%	5
CO2	Quiz 2 + Assignment	50% + 50%	5
CO3	Quiz 3 + Class Test	40% + 60%	5
CO4	Quiz 4 + Class Test	40% + 60%	5
CO5	Quiz 5 + Assignment	50% + 50%	5
	Attendance	100%	5
Total Marks			30
Assessment Plan for External Component			

Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term – I	CO1 (10 marks) + CO2 (5 marks)	15
Mid Term -II	CO2 (5 marks) + CO3 (10 Marks)	15
End Term	CO1+CO2+CO3+CO4+CO5 (4+4+4+14+14 Marks)	40
Total Marks		70

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	3	Advanced Data Structure	25CS303	3	0	0	3
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL				
36	0	0	54*		90		

** Participation in online coding contests on popular coding platforms. And NPTEL certification in any relevant course (if available).*

Course Objective (s)		
The course aims:		
1. Develop a strong foundation in fundamental data structures such as stacks, queues, trees, heaps, graphs, and tries, including their representations, operations, and efficient implementation for data organization and problem-solving.		
2. Build proficiency in algorithmic problem-solving through recursion, divide-and-conquer techniques, recurrence relations, time complexity analysis, and the design and implementation of searching, sorting, tree, and graph algorithms.		
3. Enhance programming skills using C++ and the Standard Template Library (STL) by implementing and applying built-in data structures and algorithms to solve practical computing problems efficiently.		
Syllabus		
Unit No.	Topics	CI Hours
1	Stack: Stack Introduction: Terminology, and representation (arrays, and linked list). Stack operations: Push, pop, peek, and others. Stack standard applications: Infix, prefix, and postfix notations, converting one into another, and expression evaluation. Queue: Queue Introduction: Terminology, types (linear, circular, dequeue, and priority queue), representation (arrays, and linked list). Queue operations: Enqueue, dequeue, peek, and others. Stack and Queue in STL: Using built-in Stack and Queue from STL.	6
2	Recursion: Recursion Introduction: Terminology, key components, types (direct and indirect, singly, multiply, tail, non-tail, and nested), and optimization, iteration and recursion trade-off. Recursion through problems: Understanding recursion through problems.	6

	Introduction to Divide and Conquer Approach. Recursive sorting algorithms (Merge sort and Quick sort). Time complexity of recursive algorithms: Recurrence relation and finding time complexity of recursive algorithms (substitution/iteration, recursion tree, and Master Theorem).	
3	Tree: Tree Introduction: Terminology and representation. Tree types (Complete Binary Tree, Strict Binary Tree, Skewed Binary Tree, Muti-way Tree). Binary Tree (BT) and Binary Search Tree (BST): Definitions, terminology and representations. Traversal (Breadth First Traversal and Depth First Traversal - Pre Post and In), and operations (create, search, delete, and insert). Expression Trees. Height Balanced Trees: Introduction, terminology, and representation of AVL Tree, B-Tree, B+ Tree, Red - Black Tree, Threaded Binary Tree. Operations on height-balanced trees (create, search, delete, and insert). Tries: Introduction, building a Trie data structure, basic trie operations (insert, search, and prefix search).	8
4	Heap: Heap Introduction: Terminology, and representation, heap property, heap types (max and min heap). Heap operations: Building a heap, operations over a heap, and maintaining a heap, heap sort. Heap as Priority Queue: Use case of Stack, Queue and Priority Queue. Priority Queue in STL: Using built-in Priority Queue from STL. Writing custom comparators.	8
5	Graphs: Graph Introduction: Terminology, and representation (adjacency matrix, adjacency list), types of graphs (directed, undirected, weighted, unweighted, cyclic, acyclic, bipartite graph, complete graph). Graph Traversal: Depth First Search (DFS), Breadth First Search (BFS). Applications of BFS and DFS: Connected Components, Cycle Detection, Topological Sorting, Directed Acyclic Graph (DAG), Strongly Connected Components, Articulation Points, Bridges, Bi-connected Components.	8

Prescribed Text Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein	4 th	The MIT Press	2022
2	Fundamentals of Computer Algorithms	Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran	2 nd	Universities Press 2008	2008

3	Data Structures, Algorithms and Applications in C++	Sartaj Sahni	2 nd	Universities Press	2005
4	Data structure using c and C++	Yedidiah Langsam, Moshe J. Augenstein	2 nd	Pearson Education	2006

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Data Structures and Algorithms Made Easy	Narasimha Karumanchi	6 th	Careermonk Publications	2016
2	Data Structures and Algorithms in C++	Adam Drozdek	4 th	Cengage Learning	2013
3	Introduction to the Design and Analysis of Algorithms	Anany Levitin	3 rd	Pearson Education	2017
4	Algorithms	Robert Sedgewick	4 th	Addison Wesley	2011

Web References

1.	NPTEL Course Link : https://nptel.ac.in/courses/106102064
2.	NPTEL Course Link : https://nptel.ac.in/courses/106105164
3.	NPTEL Course Link : https://nptel.ac.in/courses/106106127
4.	NPTEL Course Link : https://onlinecourses.nptel.ac.in/noc26_cs08/preview

Course Outcomes (COs):

Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Understand and apply stack and queue data structures to solve computational problems efficiently.	K3
CO2	Understand recursion to solve computational problems, including the implementation of recursive algorithms such as sorting.	K3

CO3	Understand, analyze and apply appropriate tree-based data structures, including height-balanced trees and tries, for efficient data organization and searching.	K4
CO4	Understand and apply heap-based data structures, including binary, binomial, and Fibonacci heaps, and utilize priority queues to develop optimal solutions.	K3
CO5	Understand and analyze graph data structures and their traversal algorithms, and apply them to develop efficient solutions.	K4

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	2	1	1	2	1					1			
CO2	3	2	1	1	2	1					1			
CO3	3	2	1	1	2	2					1			
CO4	3	2	1	1	2	2					1			
CO5	3	2	1	1	2	2					1			
Course	3.00	2.00	1.00	1.00	2.00	1.60					1.00			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1 + CO2	Coding Contest	100%	7
CO3 + CO4	Coding Contest	100%	8
All COs	Coding Contest	100%	10
Attendance		100%	5
Total Marks			30

Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term - I	CO1 (10 marks) + CO2 (5 marks)	15
Mid Term -II	CO2 (5 marks) + CO3 (10 marks)	15
End Term	CO1 + CO2 + CO3 + CO4 + CO5 (4 + 4 + 4 + 14 + 14 marks)	40
Total Marks		70

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	III	Data Visualization using Python	25VA301	2	0	2	3
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
24	-	24	42		90		

Course Objective (s)		
The course aims:		
1. Understand principles and rationale of effective data visualization.		
2. Learn Python libraries and tools for visualization tasks.		
3. Select and build appropriate visual representations for different data types.		
4. Integrate visualization techniques with real-world datasets.		
5. Prepare students to transition into advanced data handling and AI workflows		
Syllabus		
Unit No.	Topics	CI Hours
1	Introduction Python: Basics (Variables, Integers, Strings, Booleans), Operators, Data- Types (Numeric, Boolean, Dictionary, Set, Sequence type). If-Else, For Loops, While Loops, User defined functions). Data Visualization: Introduction to Data Visualization, Need for Data Visualization, Industry Use Cases, Human Perception & Visual Encoding, Types of Charts, real-world datasets, Data Formats (CSV, JSON, Excel), Python for Visualization (IDE, Jupyter, VS Code, kaggle), Data Handling with Pandas & NumPy Hands-On: Setting up Jupyter/Anaconda and performing basic EDA on a "Dirty" dataset to find missing values.	9
2	Foundation for Visualization Plot structure: figure, axes, labels, Line plots: single and multi-series, Bar charts and comparisons, Scatter plots, markers & styles, Histograms: distribution visualisation, Subplots & layout management, Custom styling & annotations, Saving/exporting publication-ready plots, Case Study: Exploratory Visualization Workflow Hands-On: Generating a multi-plot "Exploratory Dashboard" using Seaborn for correlation analysis	9
3	Statistical Visualization Introduction to Seaborn aesthetics, Categorical plots, Distribution plots, Sampling Techniques Measure of Central tendency, Measure of Dispersion, Outliers, Correlation (Pearson and Spearman), Covariance, Box plot, Probability Density Functions, Regression, Case Study Hands-On: Building a "Global Migration" interactive map using Folium/Plotly.	10

4	Interactive Visualization Need for interactive visuals, Plotly basics: scatter, bar & line, Interactive features: hover, click, zoom, Animation & time-series visuals, Dash framework fundamentals, Layouts & callback functions, Packaging dashboards for web, feature storytelling, Streamlit, real-time dashboards, APIs, D3.js, Case: Interactive visualization of business data Hands-On: Deploying a live Python-based dashboard on a local server or GitHub Pages. Exposure to Web-based Visualization (D3.js & SVG): advanced web-based visualization techniques without increasing complexity.	10
5	Visualization Best Practices & Real Case Studies Visual best practices & ethics, Color theory & accessibility, Telling stories with data, Visualizing time-series & geo-data, Interactive vs static graphs, Business Intelligence reports using Power BI, Review of dashboards integrating multiple charts, portfolio, GenAI usage, capstone project Hands-On: Visualizing the performance of a simple classifier (Confusion Matrix, Feature Importance) to bridge the gap toward the AI syllabus. Deployment & Portfolio Development: Deployment (Streamlit Cloud / GitHub) to create live projects. GenAI Tools: AI-assisted data analysis (e.g., using GPT for EDA insights and visualization code generation)	10

Prescribed Text Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Python Data Science Handbook: Essential Tools for Working with Data	Jake VanderPlas	1 st Edition	O'Reilly Media	2016
2	Storytelling with Data: A Data Visualization Guide for Business Professionals	Cole Nussbaumer Knaflitz	1 st Edition	Wiley	2015
3	Data Visualization with Python and JavaScript	Kyran Dale	2 nd Edition	O'Reilly Media	2022
4	Python for Data Analysis	Wes McKinney	3 rd Edition	O'Reilly Media	2022
5	Interactive Data Visualization for the Web	Scott Murray	2 nd Edition	O'Reilly Media	2017
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Python Data Science Handbook: Essential Tools for Working with Data	Jake VanderPlas	2 nd Edition	O'Reilly Media	2022
2	Python Data Structures and Algorithms	Benjamin Baka	1 st Edition	Packt Publishing	2017

3	Storytelling with Data: A Data Visualization Guide for Business Professionals	Cole Nussbaumer Knaflitz	10 th Anniversary Edition	John Wiley & Sons	2025
4	Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures	Claus O. Wilke	1 st Edition	O'Reilly Media	2019
5	Practical Statistics for Data Scientists: 50+ Essential Concepts Using R and Python	Peter Bruce, Andrew Bruce, and Peter Gedeck	2 nd Edition	O'Reilly Media	2020
6	Data Science and Data Analytics Using Python	Munesh Chandra Trivedi, Anil Kumar Dubey	1 st Edition	Khanna Publishing House	2025

Web References

1.	https://nptel.ac.in/courses/106107220
2.	https://nptel.ac.in/courses/106106145
3.	https://www.kaggle.com/learn/data-visualization
4.	https://github.com/jakevdp/pythondatasciencehandbook
5.	https://plotly.com/python/

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Understand core concepts of python, data visualization and principles of visual design.	K2
CO2	Apply Python libraries to create basic and advanced visualizations.	K3
CO3	Analyze datasets and select visualization techniques for insights.	K4
CO4	Design visualization workflows for real datasets.	K5
CO5	Evaluate visualizations for clarity, accuracy and communication.	K5

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11
CO1	3	2	1	1	2	-	-	1	-	1	-
CO2	2	3	2	2	3	-	-	1	1	2	-
CO3	2	3	3	3	2	1	-	1	1	2	-
CO4	2	2	3	3	3	1	-	1	2	3	2
CO5	2	3	2	3	2	1	1	2	2	3	1
Course	2.2	2.6	2.2	2.4	2.4	1	1	1.2	1.5	2.2	1.5

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz, Debugging	50% + 50%	5
CO2	Quiz, Debugging	50% + 50%	5
CO3	In house competition participation	100%	5
CO4	Debugging, competition participation	30% + 70%	5
CO5	Debugging, competition participation	30% + 70%	5
	Attendance	100%	5
Total Marks			30
Assessment Plan for External Component			
Assessment Tool		COs Coverage	Weightage of Marks in External Component
Mid Term - I		CO1 (10 marks) + CO2 (5 marks)	15
Mid Term -II		CO2 (5 marks) + CO3 (10 Marks)	15
End Term		CO1+CO2+CO3+CO4+CO5 (4+4+4+14+14 Marks)	40
Total Marks			70

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
BTECH	III	Fundamental Cloud Computing	2VA302	2	0	2	3
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
24	0	24	42		90		

Course Objective (s)		
The course aims:		
1. Understand the fundamental concepts, service models, and deployment models of cloud computing.		
2. Explain virtualization, cloud infrastructure, networking, storage, and computing concepts used in cloud environments.		
3. Apply cloud security, identity management, and governance principles for secure cloud adoption.		
4. Understand cloud resource management, monitoring, cost optimization, and cloud architecture best practices.		
5. Gain practical exposure to cloud services using any leading cloud platform (AWS, Microsoft Azure, or Google Cloud Platform).		
Syllabus		
Unit No.	Topics	CI Hours
1	Introduction to Cloud Computing Evolution of Computing, Introduction to Cloud Computing, Characteristics and Benefits of Cloud Computing, Cloud Service Models (IaaS, PaaS, SaaS), Cloud Deployment Models (Public, Private, Hybrid, Community), Cloud Infrastructure Components (Compute, Storage, Networking), Overview of Major Cloud Providers, Applications of Cloud Computing.	9
2	Virtualization and Cloud Management Introduction to Virtualization, Types of Virtualization, Hypervisors, Virtual Machines, Introduction to Containerization, Resource Pooling, Elasticity and Scalability, Cloud Resource Provisioning, Introduction to Cloud Compute Services, Amazon EC2 (Elastic Compute Cloud) –Launching and Managing Virtual Machine Instances, Cloud Cost Models and Pricing Basics, Case Study on Cloud Adoption. Lab: Creating and Managing Virtual Machine Instances using AWS EC2	9
3	Cloud Networking and Compute Services Basics of Cloud Networking, Public and Private Networks, IP Addressing and Subnetting, Routing, Firewalls, Domain Name System (DNS), Virtual Compute Resources, Load Balancing, Auto Scaling Concepts, Introduction to Cloud Monitoring. Lab: Creating and Configuring Amazon VPC (Virtual Public Cloud) with Public and Private Subnets.	10
4	Cloud Storage and Database Services Cloud Storage Fundamentals, Object Storage, Block Storage, File Storage, Storage Selection Criteria, Data Backup and Recovery, Introduction to Amazon S3 (Simple Storage Service), Relational Databases, NoSQL Databases, Introduction to Data Warehousing. Lab: Creating and Managing Storage Buckets using Amazon S3, Database Deployment, Backup and Recovery Exercise.	10

5	Cloud Security and Best Practices Cloud Security Fundamentals, Shared Responsibility Model, Identity and Access Management (IAM), Authentication and Authorization, Encryption Basics, High Availability Concepts, Cost Optimization Basics, Cloud Monitoring, Cloud Best Practices, Case Studies on Cloud Security and Cloud Adoption	10
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Prescribed Text Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Cloud Computing: Concepts, Technology & Architecture	Thomas Erl, Ricardo Puttini, Zaigham Mahmood	1st Edition	Pearson Education	2013
2	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg, Andrzej Goscinski	1st Edition	Wiley	2013
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	AWS for Solutions Architects	Saurabh Shrivastava, Neelanjali Srivastav	2nd Edition	Packt Publishing	2020
Web References					
1.	https://docs.aws.amazon.com/				
2.	https://aws.amazon.com/training/				
3.	https://learn.microsoft.com/en-gb/azure/				

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Explain the basic concepts, service models, deployment models, and infrastructure of cloud computing.	K2
CO2	Apply virtualization, networking, and compute concepts to manage cloud resources.	K3
CO3	Demonstrate the use of cloud storage, databases, and resource management techniques for cloud applications.	K3
CO4	Apply cloud security, identity management, and monitoring principles to build secure cloud environments.	K4
CO5	Evaluate cloud architecture, cost optimization, governance, and best practices for efficient cloud adoption.	K5

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO1	3	2	1	2		2	2	2	2		2			

CO2	1	1	3		2	1		2		2	1			
CO3	1	2		2	3	1	3		1		1			
CO4	2		2		2				3	1	3			
CO5	2	1		1			2			2	1			
Course	1.8	1.5	2	1.6	2.3	1.3	1.4	2	2	1.6	1.6			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz, Debugging	50% + 50%	5
CO2	Quiz, Debugging	50% + 50%	5
CO3	In house competition participation	100%	5
CO4	Debugging, competition participation	30% + 70%	5
CO5	Debugging, competition participation	30% + 70%	5
	Attendance	100%	5
Total Marks			30
Assessment Plan for External Component			
Assessment Tool		COs Coverage	Weightage of Marks in External Component
Mid Term - I		CO1 (10 marks) + CO2 (5 marks)	15
Mid Term -II		CO2 (5 marks) + CO3 (10 Marks)	15
End Term		CO1+CO2+CO3+CO4+CO5 (4+4+4+14+14 Marks)	40
Total Marks			70

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	III	Object Oriented Programming System Lab	25CS351	0	0	2	1
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
0	0	24	06		30		

Course Objective (s)		
The course aims:		
1. Develop programming skills using classes, functions, and modern C++ language constructs for solving computational problems.		
2. Provide practical experience in object creation, constructor mechanisms, memory management, and smart pointer-based resource handling.		
3. Enable students to implement compile-time polymorphism and operator overloading techniques for developing reusable software components.		
4. Develop the ability to design and implement object-oriented solutions using inheritance, runtime polymorphism, and code reusability principles.		
5. Introduce generic programming concepts and Standard Template Library (STL) to build efficient, reusable, and maintainable software applications.		
Syllabus		
Exp. No.	List of Experiments	LI Hours
1	Design a program to convert a procedural implementation of a student record system into an object-oriented approach using classes and objects.	1
2	Develop a program to demonstrate parameter passing techniques (call by value, reference, and address) and analyze their effects on variable modification.	1
3	Implement a program to demonstrate inline functions, default arguments, and function overloading for performing arithmetic operations.	1
4	Write a program using modern C++ features such as auto and range-based for loop to traverse and display elements of a collection.	1
5	Design a class to represent a bank account with proper data hiding and member functions for deposit and withdrawal operations.	1
6	Develop a program to demonstrate different types of constructors and destructor behavior in object lifecycle management.	1
7	Implement a program using static members and friend functions to illustrate shared data and controlled access.	1
8	Write a program to pass objects as arguments and return objects from functions to perform operations on user-defined data.	1
9	Design a program using pointers to objects and arrays of objects to manage dynamic data structures.	1

10	Implement a program using smart pointers (unique_ptr, shared_ptr) to manage dynamic memory safely and avoid memory leaks.	1
11	Develop a program to demonstrate function overloading for performing operations on different data types.	1
12	Design a program to overload unary and binary operators for a user-defined class (e.g., complex numbers).	1
13	Develop a program to demonstrate type conversion between basic data types and user-defined classes.	1
14	Implement a program demonstrating single and multilevel inheritance using a real-world example.	1
15	Design a program to illustrate function overriding in derived classes.	1
16	Develop a program using virtual functions to demonstrate runtime polymorphism with base class pointers.	1
17	Implement an abstract class with pure virtual functions and derive classes to provide specific implementations.	1
18	Design a system using composition and compare it with inheritance-based design for the same problem.	1
19	Develop a function template to perform generic operations such as swapping or sorting of elements.	1
20	Implement a class template to create a generic stack or queue data structure.	1
Exp. No.	Value Added Experiments	LI Hours
21	Write a program using STL vector and iterators to store and process a list of elements.	1
22	Design a program using STL map or set to manage key-value data efficiently.	1
23	Implement STL algorithms such as sort() and find() to perform operations on collections.	1
24	Develop a program using lambda expressions to customize sorting or filtering operations.	1

Prescribed Textbooks					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Object Oriented Programming with C++	Reema Thareja	Revised 1 st Ed	Oxford University Press	2018
2	Mastering C++	K R Venugopal, Rajkumar, T Ravishankar	25 th Reprint	McGraw Hill Education	2006
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year

1	A Tour of C++	Bjarne Stroustrup	3	Addison-Wesley Professional	2022
2	C++ Primer	Stanley B. Lippman, Stanley Lippman , Barbara Moo	5	Addison-Wesley Professional	2013
Web References					
1. https://onlinecourses.nptel.ac.in/noc25_cs34					
2. https://onlinecourses.nptel.ac.in/noc21_cs38					

***Certification for the self-learning component should preferably be included in the assessment.**

Course Outcomes (COs):		
Upon the completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Implement basic programs using classes, functions, and modern C++ constructs	K3
CO2	Develop applications using constructors, memory management, and smart pointers	K3
CO3	Apply compile-time polymorphism and operator overloading in problem-solving	K4
CO4	Design programs using inheritance and runtime polymorphism concepts	K4
CO5	Build generic and reusable solutions using templates and STL	K5

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	1				1		1							
CO2	1						1							
CO3	2		1				1							
CO4			3				1							
CO5			1		1		1	2	1	1				
Course Avg.	1.3		1.6		1.00		1	2.00	1.00	1				

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern

Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	100

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Quiz 1 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO2			
	Attendance		100%	5
PA2	CO3	Quiz 2 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO4			
	CO5			
	Attendance		100%	5
Total Marks				50
Assessment Plan for External Component				
Assessment Tool			COs Coverage	Weightage of Marks in External Component
End Term			All COs	50
Total Marks				50

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	III	Operating System Lab	25CS352	0	0	2	1
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
0	0	24	06		30		

Course Objective (s)		
The course aims:		
1.	To provide hands-on experience with operating system environments, commands, and shell scripting.	
2.	To develop programs for process management, CPU scheduling, and deadlock handling techniques.	
3.	To implement memory management and page replacement algorithms for performance analysis.	
Syllabus		
Exp. No.	List of Experiments	LI Hours
1	Study of software and hardware requirements of various Operating Systems (Windows/Linux/Unix) and Implementation of DOS commands on command prompt.	2
2	Shell Scripting	2
3	Implementation of Process and thread (Life cycle of process): (i) Process creation and Termination; (ii) Thread creation and Termination	2
4	Write a program to implement the FCFS CPU Scheduling Algorithm.	2
5	Write a program to implement the SJF CPU Scheduling Algorithm.	2
6	Write a program to implement the SRTF CPU Scheduling Algorithm.	2
7	Write a program to implement Priority (Primitive and Non-primitive) CPU Scheduling Algorithm.	2
8	Write a program to implement the Round Robin CPU Scheduling Algorithm	2
9	Write a program to simulate Banker’s Algorithm.	2
10	Write a program to simulate internal fragmentation according to First fit, Best Fit, and Worst Fit.	2
Exp. No.	Value Added Experiments	LI Hours
11	Write a program to simulate the FIFO Page Replacement Algorithm	2

12	Write a program to simulate the LRU Page Replacement Algorithm	2
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Prescribed Textbooks					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Operating System Concepts	Silberschatz, Abraham, Peter B. Galvin, and Greg Gagne.	10th	John Wiley & Sons	2018
2	Modern Operating Systems	Andrew S. Tanenbaum	5th Edition	Pearson	2022
3	Operating Systems: Internals and Design Principles	William Stallings	10th Edition	Pearson	2018
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Modern Operating Systems	Andrew S. Tanenbaum	5th Edition	Pearson	2022
Web References					
1. NPTEL: Operating System Fundamentals :By Prof. Santanu Chattopadhyay https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs156 .					
2. Udemy Course: The Complete Operating Systems Course: From Zero to Hero! https://www.udemy.com/course/the-complete-operating-systems-course-from-zero-to-expert/?couponCode=UDEAFFHP22025 .					

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs):		
Upon the completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Understand the basic hardware and software requirements of operating system and implementation of basic DOS commands and shell scripting.	K3
CO2	Analyze and simulate various CPU scheduling algorithms for process management.	K4
CO3	Analyze and implement file storage allocation techniques and memory management schemes	K4
CO4	Implement algorithms for deadlock avoidance and safe sequence .	K3
CO5	Analyze the performance of various page replacement algorithms.	K4

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	2	1	3			1	1	1	2		3			
CO2	2	3	3	3	3	1	1	1	2		3			
CO3	2	3	3	3	3	1	1	1	2		3			
CO4	2	3	3	3	3	1	1	1	2		3			
CO5	2	3	3	3	3	1	1	1	2		3			
Course Avg.	2.00	2.60	3.00	2.40	2.40	1.00	1.00	1.00	2.00		3.00			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	100

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Quiz 1 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO2			
	Attendance		100%	5
PA2	CO3	Quiz 2 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO4			
	CO5			
	Attendance		100%	5
Total Marks				50
Assessment Plan for External Component				
Assessment Tool			COs Coverage	Weightage of Marks in External Component
End Term			All COs	50
Total Marks				50

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	3	Advanced Data Structure Lab	25CS353	0	0	2	1
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL				
0	0	24	06*		30		

* Participation in online coding contests on popular coding platforms.

Course Objective (s)		
The course aims		
1. The objective of this course is to develop a strong foundation in fundamental data structures and algorithms, including stacks, queues, recursion, trees, heaps, and graphs.		
2. Students will learn to implement, debug, and apply these techniques to solve computational problems efficiently.		
3. To build competency in designing and implementing recursive, iterative, and algorithmic solutions using appropriate data structures.		
Syllabus		
Exp. No.	List of Experiments	LI Hours
1	Given a stack of size N, implement PUSH and POP operations following the LIFO (Last In First Out) principle. Handle overflow and underflow conditions, and display the final contents of the stack. Also, given an infix expression, convert it into its equivalent postfix expression using a stack by following operator precedence and associativity rules. Finally, evaluate the generated postfix expression using a stack and display the computed result.	2
2	Given a queue of size N, implement ENQUEUE and DEQUEUE operations using an array or linked list while following the FIFO (First In First Out) principle. Handle overflow and underflow conditions, and display the final state of the queue. Also, implement a circular queue using an array, where vacant positions created by deletions are reused by maintaining the circular property. Perform the given sequence of operations and display the final state of the circular queue.	2
3	Given an integer N, compute the factorial and the Nth Fibonacci number using recursion and related problems based on recursion without using any iterative constructs.	2
4	Given an array of N integers, sort the array using either Merge Sort or Quick Sort based on the Divide and Conquer approach, and print the sorted array.	2
5	Given a binary tree with N nodes, perform Preorder, Inorder, Postorder, and Level Order traversals, and display the traversal sequences.	2

6	Given a Binary Search Tree (BST), implement INSERT, SEARCH, and DELETE operations while maintaining BST properties, and display the inorder traversal after all operations. Also, given a set of strings, implement a Trie supporting INSERT, SEARCH, and PREFIX SEARCH operations, and display the result of each query.	2
7	Given an array of N integers, build a Min Heap or Max Heap and implement INSERT, DELETE, and PEEK operations while maintaining the heap property. Display the final state of the heap after performing all operations.	2
8	Given an array of N integers, sort the array using the Heap Sort algorithm by first building a heap and then repeatedly extracting the root element to obtain the sorted sequence. Also, implement a Priority Queue using a Heap or STL priority_queue, supporting INSERT, DELETE (highest-priority element), and TOP/PEEK operations, and display the result of each operation.	2
9	Given a graph with N vertices and M edges, represent the graph using an adjacency matrix or an adjacency list. Perform Breadth First Search (BFS) and Depth First Search (DFS) starting from a given source vertex, and print the traversal order. Also, determine the number of connected components and detect whether the graph contains a cycle using BFS or DFS.	2
10	Given a directed graph, determine whether it is a Directed Acyclic Graph (DAG). If the graph is acyclic, perform Topological Sorting and print the ordering of the vertices; otherwise, report that topological sorting is not possible.	2
Exp. No.	Value Added Experiments	LI Hours
11	Develop a program that accepts an arithmetic expression in infix notation, converts it to postfix notation using a stack, and evaluates the postfix expression. The program should support operators (+, -, *, /, %, ^), parentheses, and multi-digit operands. Display both the postfix expression and the final evaluated result.	2
12	Given a graph represented using an adjacency list, implement BFS and DFS traversals. Using BFS, find the shortest path (minimum number of edges) between a given source and destination vertex. Also, use DFS to detect whether the graph contains a cycle and determine the number of connected components.	2

Prescribed Text Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein	4 th	The MIT Press	2022
2	Data Structures, Algorithms and Applications in C++	Sartaj Sahni	2 nd	Universities Press	2005
Reference Books					

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Data Structures and Algorithms Made Easy	Narasimha Karumanchi	6 th	Careermonk Publications	2016
2	Data Structures and Algorithms in C++	Adam Drozdek	4 th	Cengage Learning	2013
Web References					
1. https://nptel.ac.in/courses/106102064					

Course Outcomes (COs):		
Upon the Completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Implement and debug stack and queue data structures to develop efficient solutions.	K4
CO2	Implement and debug recursive algorithms, including recursive sorting techniques, to solve computational problems.	K4
CO3	Implement and debug tree-based data structures, including height-balanced trees and tries, for efficient data organization and searching.	K4
CO4	Implement and debug heap-based data structures by utilizing priority queues to develop efficient solutions.	K4
CO5	Implement and debug graph data structures and traversal algorithms and apply them to diverse applications.	K4

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	2	2	2	3	1			1		2			
CO2	3	2	2	2	3	1			1		2			
CO3	3	2	2	2	3	1			1		2			
CO4	3	2	2	2	3	1			1		2			
CO5	3	2	2	2	3	1			1		2			
Course	3.00	2.00	2.00	2.00	3.00	1.00			1.00		2.00			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern			
Internal Component (CIA)		External Components	Total Marks
PA1	PA2		
25	25	50	100

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Record of programs written during labs + Coding Contest.	20% + 80%	20
	CO2			
	Attendance		100%	5
PA2	CO3	Record of programs written during labs + Coding Contest.	20% + 80%	20
	CO4			
	CO5			
	Attendance		100%	5
Total Marks				50
Assessment Plan for External Component				
Assessment Tool		COs Coverage	Weightage of Marks in External Component	
Coding Contest (30%) + Viva (20%)		All COs	50	
Total Marks			50	

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	III	FSD Workshop-I	25VA351	0	0	2	1
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
0	0	24	06		30		

Course Objective (s)		
The course aims:		
1. To introduce students to server-side development concepts including the architecture, core modules, and event-driven programming model of Node.js.		
2. To familiarize students with asynchronous programming techniques and file system operations for efficient handling of files and directories in Node.js		
3. To enable students to develop web servers and implement routing mechanisms using HTTP modules and Express.js.		
4. To provide knowledge of NoSQL database concepts and data modeling using MongoDB for performing CRUD operations.		
Syllabus		
Exp. No.	List of Experiments	LI Hours
1	1. Create a custom EventEmitter that triggers "greet" or "exit" 2. Simulate DOM-like event handling in Node.js using events 3. Visualize the event loop using setTimeout, setImmediate, and process.nextTick	2
2	Perform CRUD operations on files using fs module	2
3	Create a basic HTTP server using http.createServer(), Respond with “Hello World” and return headers + status code	2
4	Set up a basic GraphQL API using or express-graphql	2
5	Create routes that handle GET, POST, PUT, DELETE in Node.js	2
6	Create a simple Express.js server that: Serves an HTML file using the fs module.Has routes like /, /about, and /contact. Each route should serve a separate HTML file.	2
7	Install MongoDB locally (or use MongoDB Atlas). Create a new database named studentDB with a collection students	2
8	Perform the following on studentDB.students collection: Insert 5 student records. Find all students older than 20. Update a student’s age using \$set. Delete one student.	2

9	Create a REST API using Express and MongoDB: POST /students – Add a new student GET /students – Retrieve all students PUT /students/:id – Update a student by ID DELETE /students/:id – Delete a student	4
10	Push your MongoDB + Express project to GitHub. Ensure: Proper README.md .gitignore included Folder structure is clean	4
Exp. No.	Value Added Experiments	LI Hours
11	Dockerize your Node.js + MongoDB app: Write a Dockerfile, Build and run the Docker image locally	2
12	Create a Student Management System(SMS) applications using react, express, and mongo DB then Dockerize and deploy on render	4

Prescribed Textbooks					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Web Technologies – A Developer's Perspective	N.P. Gopalan and J. Akilandeswari	3rd	Pearson	2025
2	Node.js, Express, MongoDB & JavaScript	William Stallings	11th	Pearson	2024
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Learning Web Design	Jennifer Robbins	5th	Mc Graw Hill	2012
2	Web Development with Node and Express	Ethan Brown	8th	"O'Reilly Media	2014
Web References					
1. AWS Academy – https://aws.amazon.com/training/ 2. Microsoft Learn – https://learn.microsoft.com 3. Google Cloud Skills Boost – https://cloudskillsboost.google 4. Coursera – https://www.coursera.org					

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs):		
Upon the completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Understand the architecture, core modules, and event-driven programming model of Node.js for building scalable server-side applications.	K2
CO2	Apply asynchronous programming concepts and Node.js file system modules to perform file and directory operations efficiently.	K3
CO3	Develop web servers and implement routing mechanisms using HTTP modules and Express.js to handle client requests and responses.	K3
CO4	Develop web servers and implement routing mechanisms using HTTP modules and Express.js to handle client requests and responses.	K3
CO5	Design, develop, and deploy RESTful APIs integrating Node.js, Express.js, and MongoDB, and test the APIs using Postman with version control using GitHub.	K6

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	2	1	-	2						1			
CO2	3	3	2	1	3						1			
CO3	3	3	3	2	3				1	1	1			
CO4	3	3	3	2	3			1	2		2			
CO5	3	2	3	2	3			1	2	2	3			
Course Avg.	3.00	2.60	2.40	1.40	2.80			0.40	1.00	0.60	1.60			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	100

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Quiz 1 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO2			
	Attendance		100%	5
PA2	CO3	Quiz 2 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO4			
	CO5			
	Attendance		100%	5
Total Marks				50
Assessment Plan for External Component				
Assessment Tool			COs Coverage	Weightage of Marks in External Component
End Term			All COs	50
Total Marks				50

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	IV	Computer Organization and Architecture	25CS401	3	0	0	3
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
36	0	0	54		90		

Course Objective (s)		
The course aims:		
1. To introduce principles of computer organization and the basic architectural concepts.		
2. To provide basic knowledge of organization, design, and programming of a simple digital computer.		
3. To make aware about topics include computer arithmetic, instruction set design, microprogrammed control unit, pipelining, memory organization and I/O systems, and multiprocessors		
Syllabus		
Unit No.	Topics	CI Hours
1	Introduction: Von Neumann and Harvard architecture, Basic of computer system and their functional unit , Bus and Types of Bus, Multiplexer, decoder, Tri state buffer, Register and their types, General register organization, Instruction format and their types, Addressing modes, Architecture diagram of 8085.	8
2	ALU: Basics Number System, Look ahead carries adders, Booth's algorithm, Division Algorithm (Restoring Method), Floating point arithmetic operation, Arithmetic & logic unit design, IEEE Standard for Floating Point Numbers	7
3	Control Unit: Fundamental Concepts of Instruction Cycles, Introduction to Pipelining, RISC and CISC, Hard-wired & Micro-Programmed Control Unit, Introduction to GPU Architecture and NPU, Comparison between GPU and CPU Architecture	7
4	Memory: Memory organization: Concept of hierarchical memory organization, Memory Interleaving, Cache Coherence: problem and solution, Cache memory, Cache size vs. block size, Cache address mapping (Direct, Fully Associative, Set Associative) & Replacement algorithms (LRU, FIFO, LFU), write policies (Write-Through, Write-Back).	7
5	IO Processor: Peripheral devices, I/O interface, I/O ports, Interrupts: interrupt hardware, types of interrupts and exceptions, Modes of Data Transfer: Programmed I/O, interrupt initiated I/O and Direct Memory Access, [Gate Contents Coverage] I/O channels and processors, Flynn's Classification SISD, SIMD, MISD, MIMD, Array processor (attached array, SIMD array), Interconnection Architecture and arbitration.	7

Prescribed Textbooks

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Computer System Architecture	Morris Mano	3rd	Pearson	2017
2	Computer System Architecture, Revised 3/e Computer System Architecture, Designing for Performance	William Stallings	11th	Pearson	2022
3	Computer Organization	Carl Hamacher, Zvonko Vranesic, Safwat Zaky	5th	Mc Graw Hill	2012
4	Fundamental of Computer Organization and Design	Sivarama P Dandamudi	3rd	Springer	2003

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	Fundamentals of Computer Organization and Architecture	Jyotsna Sengupta	1st	Deep & Deep Publications	2009
2	Computer Architecture and Organization	John P. Hayes	3rd	TMH	1998
3	Computer Architecture	Behrooz Parahami	8th	Oxford Univ. Press	2011

Web References

1. NPTEL Course Link: <https://nptel.ac.in/courses/106106166>

2. Books for COA: <https://www.speedstar.co.in/computer-books-on-coa/>

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs):

Upon the completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Study and understand the fundamental concepts of computer organization and architecture.	K2
CO2	Understand and design of arithmetic & logic unit.	K3
CO3	Understand Control Unit design and Architectures with their differences.	K3
CO4	Understand the concept of memory organization and operations.	K3
CO5	Understand importance of interface, differentiate different mode of transfer and classify parallel machine models.	K3

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	3	3	1							1			1
CO2	3	3	3	3							2	1		
CO3	2	2	2	2							1			1
CO4	3	3	3	2							2	1		
CO5	2	2	1	2							1			1
Course Avg.	2.60	2.60	2.40	2.00							1.40	1.00		1.00

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz 1	100%	5
CO2	Quiz 2 + Assignment	50% + 50%	5
CO3	Quiz 3 + Class Test	40% + 60%	5
CO4	Quiz 4 + Class Test	40% + 60%	5
CO5	Quiz 5 + Certification Course	30% + 70%	5
	Attendance	100%	5
Total Marks			30
Assessment Plan for External Component			
Assessment Tool		COs Coverage	Weightage of Marks in External Component
Mid Term – I		CO1 (10 marks) + CO2 (5 marks)	15
Mid Term -II		CO2 (5 marks) + CO3 (10 Marks)	15
End Term		CO1+CO2+CO3+CO4+CO5 (4+4+4+14+14 Marks)	40
Total Marks			70

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech.	IV	Database Management System	25CS402	3	0	0	3
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
36	0	0	54		90		

Course Objective (s)		
The course aims:		
To understand the fundamental concepts of database systems, data models, and database design using Entity-Relationship (ER) modeling.		
2. To develop proficiency in relational database concepts, SQL queries, and normalization techniques for efficient data storage and retrieval.		
3. To learn transaction management and concurrency control mechanisms to ensure data consistency, integrity, and reliability in database systems.		
Syllabus		
Unit No.	Topics	CI Hours
1	Introduction to Database & ER Modeling Introduction: Data vs Information, Database, DBMS, File System vs DBMS, Instances & Schemas, View of Data (Three-Schema Architecture), Data Independence, Data Models: Record Based Data Models, Object Based Data Models, Database Languages, Database Application Architecture, Database Structure. ER Model: Entities, Entity Sets, Attributes, Attribute Types: Simple & Composite, Single-Values & Multi-Valued, Derived, Entity Types: Strong, Weak, Relationship Sets, Mapping Cardinalities, Participation Constraints: Total, Partial, Keys: Super Key, Candidate, Primary, Alternate, Foreign Key. Enhanced ER: Generalization, Specialization, and Aggregation. Reduction of ER Diagrams to Tables.	8
2	Relational Data Model & Relational Algebra Relational Concepts: Relational Data Model Concepts, Domains, Tuples, Relations, Integrity Constraints: Entity Integrity, Referential Integrity, Domain Integrity. Database Query Languages: Procedural Query Language, Non-procedural Query Language Relational Algebra: Fundamental operations: Select, Project, Set Union, Set Intersection, Set Difference, Cartesian Product, Rename, Joins, Division, Assignment. Extended Operations: Generalized Project, Aggregate Functions.	8
3	Structured Query Language (SQL) Introduction: Data Types, SQL Statements: DDL, DML, DCL, TCL DDL & DML: Schema Definition in SQL: Create, Alter, Drop, Truncate, rename. Basic structure of SQL Query: Select, From, Where, String Operators, Ordering the Display of Tuples: Order by, Insert, Update, Delete. Querying: Set Operations: Union, Union All, Intersect, Minus, Aggregate Functions, Null Values, Subqueries: Single Row, Multiple Row, Nested, Correlated, Group by, Having clause. Joins: Inner Join, Outer Joins: Left Outer Join, Right Outer Join, Full Outer Join, Self Joins. Join Conditions: Natural, ON, USING.	8
4	Database Design & Normalization	6

	Functional Dependencies: Introduction, Types of FDs: Trivial, Non Trivial, Transitive, Multivalued, Armstrong's Axioms, Closure of attributes, Closure of a set of FDs, Finding Super Keys and Candidate Keys using Attribute Closure, Equivalence of Set of FDs, Canonical cover. Normal Forms: Relational Schema Design Issues: Insertion Anomaly, Updation Anomaly, Deletion Anomaly, 1NF, 2NF, 3NF, BCNF (Boyce-Codd Normal Form). Lossless Join Decomposition, Dependency Preserving Decomposition. Advanced Normalization: Multi-valued dependencies and 4NF, Join dependencies and 5NF.	
5	Transaction & Concurrency Control Transactions: Transaction Concept, ACID Properties, Transaction States. Concurrency: Concurrent Executions, Serializability: Conflict Serializability, View Serializability, Testing for Serializability: Precedence Graph Method, Topological Sorting, Lock-based protocols (2-Phase Locking), Timestamp-based protocols, Deadlock handling. Recovery: Recoverability, Recoverable Schedules, Cascadeless Schedules, Log-based recovery: Deferred Database Modification, Immediate Database Modification, Checkpoints.	6

Prescribed Textbooks					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Database Concepts	Korth, Silbertz, Sudarshan	7th	McGraw Hill	2019
2	Fundamentals of Database Systems	Elmasri, Navathe	7th	Addision Wesley	2015
3	An Introduction to Database Systems	Date C J	8th	Addision Wesley	2006
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Database Management Systems	RAMAKRISHNAN	3rd	McGraw Hill	2002
2	An Introduction to Database Systems	Bipin C. Desai	Revised	Galgotia Publications	2010
Web References					
1. Data Base Management System – IIT Kharagpur (Prof. Partha Pratim Das, Prof. Samiran Chattopadhyay) https://onlinecourses.nptel.ac.in/noc26_cs39/preview					
2. Database Management System Part-1 https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview					

***Certification for the self-learning component should preferably be included in the assessment.**

Course Outcomes (COs):		
Upon the completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL

CO1	Explain the fundamental concepts of database systems, data models, and ER modeling for conceptual database design.	K3
CO2	Apply relational data model concepts and relational algebra operations to represent and manipulate data in databases.	K3
CO3	Construct and execute SQL queries using DDL, DML, joins, nested queries for database operations.	K4
CO4	Apply functional dependency concepts and normalization techniques to design efficient and anomaly-free database schemas.	K4
CO5	Analyze transaction processing, concurrency control mechanisms, and recovery techniques to maintain database consistency and reliability.	K3

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	2	1	1	2			2	3		2			
CO2	3	3	2	2	2	2		3	3		2			
CO3	3	3	3	2	3	2		3	2		2			
CO4	3	3	3	2	2	2		1	1		2			
CO5	3	2	1	2	1	-		1	1		2			
Course Avg.	3.00	2.60	2.00	1.80	2.00	1.20		2.00	2.00		2.00			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Assignment/Class Test	Handwritten Assignments and surprise class test	5
CO2	Hackerrank Basic Certification in SQL		5
CO3	Hackerrank Intermediate Certification in SQL		5
CO4	Assignment/Class Test	Handwritten Assignments and surprise class test	5
CO5	Assignment/Class Test	Handwritten Assignments and surprise class test	5
	Attendance		5

Total Marks		30
Assessment Plan for External Component		
Assessment Tool	COs Coverage	Weightage of Marks in External Component
Mid Term – I	CO1 (10 marks) + CO2 (5 marks)	15
Mid Term -II	CO2 (5 marks) + CO3 (10 Marks)	15
End Term	CO1+CO2+CO3+CO4+CO5 (4+4+4+14+14 Marks)	40
Total Marks		70

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	IV	Design and Analysis of Algorithm	25CS403	3	0	0	3
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL				
36	0	0	54*		90		

** Participation in online coding contests on popular coding platforms. And NPTEL certification in any relevant course (if available).*

Course Objective (s)		
1. The course aims to develop the ability to analyze computational, combinatorial, and optimization.		
2. It emphasizes efficient problem solving through algorithmic thinking, complexity analysis, and systematic solution design.		
3. The course also enables students to evaluate the applicability of different approaches and construct efficient solutions for a wide range of computing problems.		
Syllabus		
Unit No.	Topics	CI Hours
1	Sliding Window: Introduction to fixed-size sliding window approach: Understanding the approach, time complexity of the approach. Fixed-length sliding window patterns: Window sum and average problems, frequency or counting problems, maximum and minimum value problems within a fixed range. Introduction to variable length sliding window approach: Understanding the approach, time complexity of the approach. Variable length sliding window patterns: Longest window problems and shortest window problems: distinct element problems, constraint-based subarray and substring problems, window counting problems, sliding window with hash map and hash set problems.	8
2	Prefix Sum: Introduction to Prefix Sum approach: Understanding the approach, including the concepts of cumulative sums and precomputation, understanding how prefix sums reduce repeated calculations, time complexity of the approach. Prefix sum patterns: Direct application of prefix sum approach, problems that require conversion to prefix sum pattern, using auxiliary data structure with prefix sum approach, multi dimensional prefix sum problems.	4

	Two Pointers: Introduction to the Two Pointers approach: Understanding the approach, including opposite-direction pointers, same-direction pointers, time complexity of the approach. Two Pointer patterns: Pointers moving in the same direction, pointers moving in the opposite direction, fast and slow pointers.	
3	Greedy Approach: Introduction to Greedy Approach: Understanding the approach, applicability of greedy approach (greedy choice property), complexity of greedy solutions over brute force solutions. Greedy patterns: Sorting and interval problems, resource allocation problems, greedy approach with priority queue. Greedy approach on Graphs: Disjoint Set Union (Union-Find), minimum spanning tree (Kruskal's and Prim's algorithms), single source shortest path algorithm (Dijkstra's Algorithm).	8
4	Backtracking Approach: Introduction to Backtracking Approach: Understanding the approach, state-space tree construction, state-space tree pruning, complexity analysis of backtracking. Backtracking Pattern: Binary choice problems, multiple-choice problems, permutation problems, grid based problems.	6
5	Dynamic Programming: Introduction to Dynamic Programming: Understanding the approach, applicability of Dynamic Programming approach (overlapping subproblems, optimal substructure), memoization (top-down approach) and tabulation (bottom-up approach), and space optimization techniques. Dynamic Programming Patterns: Binary choice problems, multiple choice problems, permutation problems, grid-based problems. Dynamic Programming on Trees. Dynamic Programming on Graphs: Single-source shortest path (Bellman-Ford), all-pairs shortest path algorithm (Floyd-Warshall), Travelling Salesman Problem.	10

Prescribed Text Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein	4th	The MIT Press	2022
2	Guide to Competitive Programming: Learning and Improving Algorithms Through Contests	Antti Laaksonen	2nd	Springer	2020
Reference Books					

S. No.	Book Title	Authors	Edition	Publisher	Year
1	The Algorithm Design Manual	Steven S. Skiena	3rd	Springer	2020
2	Data Structures & Algorithms	Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman	1st	Pearson Education	2002
3	Fundamentals of Computer Algorithms	Ellis Horowitz, Sartaj Sahni and Sanguthevar Rajasekaran	2nd	Universities Press	2008
4	Algorithm Design	Jon Kleinberg and Éva Tardos	1st	Pearson Education	2013
Web References					
1. NPTEL Course Link : https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs146					
2. NPTEL Course Link : https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs110					
3. NPTEL Course Link : https://onlinecourses.nptel.ac.in/noc26_cs08/preview					

Course Outcomes (COs):		
Upon the Completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Analyze sequential computation problems and apply Sliding Window techniques for efficient solutions.	K3
CO2	Analyze linear computation problems and apply Prefix Sum and Two Pointer techniques to solve problems efficiently.	K3
CO3	Analyze optimization problems and apply Greedy techniques to write efficient solutions to such problems.	K4
CO4	Understand and analyze combinatorial and constraint satisfaction problems and apply Backtracking technique to write efficient solutions.	K4
CO5	Analyze optimization problems and apply Dynamic Programming techniques to solve such problems efficiently.	K4

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	2	2	2	2						1			
CO2	3	2	2	2	2						1			
CO3	3	2	2	2	2						1			
CO4	3	2	2	2	2						1			
CO5	3	2	2	2	2						1			
Course	3.00	2.00	2.00	2.00	2.00						1.00			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1 + CO2	Coding Contest	100%	7
CO3 + CO4	Coding Contest	100%	8
All COs	Coding Contest	100%	10
Attendance		100%	5
Total Marks			30
Assessment Plan for External Component			
Assessment Tool		COs Coverage	Weightage of Marks in External Component
Mid Term - I		CO1 (10 marks) + CO2 (5 marks)	15
Mid Term -II		CO2 (5 marks) + CO3 (10 marks)	15

End Term	CO1 + CO2 + CO3 + CO4 + CO5 (4 + 4 + 4 + 14 + 14 marks)	40
Total Marks		70

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	IV	Data Handling and AI	25VA401	2	0	2	3
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
24	-	24	42		90		

Course Objective (s)			
The course aims:			
1. Understand principles and techniques for handling, processing, and transforming real-world data.			
2. Learn modern AI foundations including machine learning, data pre-processing, and model evaluation.			
3. Apply data handling pipelines and AI algorithms using open-source datasets and competition notebooks.			
4. Gain insights into scalable data solutions and current AI industry trends (MLOps, AutoML).			
5. Evaluate and visualize model results for decision-making with ethical AI considerations.			
Syllabus			
Unit No.	Topics		CI Hours
1	Introduction to Data Handling Theory: Data Lifecycle, Types & Sources (structured, unstructured), Data Quality, Data Cleaning Principles, Data Storage Systems: Files, RDBMS, NoSQL overview, Data Ingestion & ETL Fundamentals, IDE, Data Wrangling, Handling Missing Values & Outliers, Data Scaling & Normalization Techniques, Introduction to Big Data Ecosystem Hands-On: Environment Setup, Data Wrangling, Data Cleaning & Pre-processing, Scaling & Normalization		9
2	Data Handling Techniques for AI Theory: Feature Engineering & Feature Selection, hypothesis testing, Handling Categorical & Text Data (Encoding), Dimensionality Reduction (PCA), Time Series & Sequence Handling, Data Pipelines & Workflow Automation, Data Versioning & Provenance, Introduction to SQL for AI (Aggregations, Joins), Querying NoSQL (MongoDB) for AI Workflows, modern cloud platforms (Snowflake / Databricks) Hands-On: Feature Engineering, Dimensionality Reduction, SQL for AI Workflows, Time Series Preparation		9
3	Fundamentals of AI & Machine Learning Theory: AI vs Machine Learning vs Deep Learning, Supervised Learning, Unsupervised Learning, Regression Analysis: Linear Regression, Classification: Logistic Regression, KNN, Decision Trees & Random Forests, Model Evaluation Metrics (Accuracy, Precision, Recall, MAE, RMSE, R2score), bias handling, Overfitting, Underfitting & Cross-Validation.		10

	Hands-On: Regression Analysis, Classification Challenges, Model Performance Evaluation, Overfitting & Validation	
4	Advanced AI Techniques Theory: Ensemble Methods: Bagging & Boosting, Support Vector Machines & Kernel Methods, Introduction to Neural Networks (Perceptron to MLP), Model Tuning & Hyperparameter Optimization, Scalable Machine Learning with Cloud Tools (AWS/Azure), Introduction to Deep Learning Frameworks (TensorFlow/PyTorch), Ethics & Bias in AI Models, Case Study: End-to-End AI Model Deployment. Hands-On: Ensemble Modelling, Neural Network Basics, Hyperparameter Tuning, Cloud AI Simulation	10
5	Data Visualization, Interpretation & Industry Trends Theory: Managing and connecting data sources, Dashboarding & BI Tools (Power BI, Tableau) and deployment report, Spatial Data Visualization, Predictive analysis and forecasting, Interpreting Model Outputs & Explainable AI (SHAP, LIME), Industry Trends in Data & AI (MLOps, AutoML), Dataiku / Apache Airflow, Project Presentation & Reporting Hands-On: Interactive Dashboarding, Explainable AI (XAI), MLOps Fundamentals, Project	10

Prescribed Text Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow	Aurélien Geron	3rd Edition	O'Reilly Media	2022
2	Python for Data Analysis	Wes McKinney	3 rd Edition	O'Reilly Media	2022
3	Data Science and Big Data Analytics: Discovering, Analyzing, Visualizing and Presenting Data	EMC Education Services	1st Edition	John Wiley & Sons	2015
4	Data Science for Business	Foster Provost and Tom Fawcett	1st Edition	O'Reilly Media	2013
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1.	Python Data Science Handbook: Essential Tools for Working with Data	Jake VanderPlas	2nd Edition	O'Reilly Media	2022
2.	Designing Data-Intensive Applications: The Big Ideas Behind Reliable, Scalable, and Maintainable Systems	Martin Kleppmann	1st Edition	O'Reilly Media	2017
3.	Introduction to Machine Learning with Python: A Guide for Data Scientists	Andreas C. Müller & Sarah Guido	1st Edition	O'Reilly Media	2016
4.	MLOps Engineering at Scale	Robert Crowe & Hannes Hapke	1st Edition	Manning Publications	2023

5.	Data Science and Data Analytics Using Python	Munesh Chandra Trivedi, Anil Kumar Dubey	1 st Edition	Khanna Publishing House	2025
Web References					
1. https://nptel.ac.in/courses/106106139					
2. https://nptel.ac.in/courses/106106179					

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs): Upon the Completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Understand data handling principles, data lifecycle, and AI/ML foundations.	K2
CO2	Apply techniques for data pre-processing, cleaning, storage and retrieval.	K3
CO3	Analyze datasets and implement core ML algorithms using notebooks.	K4
CO4	Design end-to-end AI/ML solutions for real-world competition problems.	K5
CO5	Evaluate model performance, explainability and ethical implications.	K5

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO3
CO1	3	2	1	1	2			1		1				
CO2	2	3	2	2	3			1	1	1				
CO3	2	3	3	3	3	1		1	1	2				
CO4	2	3	3	3	3	1		1	2	3	2			
CO5	2	3	2	3	2	2	1	3	2	3	1			
Course	2.2	2.8	2.2	2.4	2.6	1.33	1	1.4	1.5	2	1.5			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz, Debugging	50% + 50%	5
CO2	Quiz, Debugging	50% + 50%	5
CO3	In house competition participation	100%	5
CO4	Debugging, competition participation	30% + 70%	5
CO5	Debugging, competition participation	30% + 70%	5
	Attendance	100%	5
Total Marks			30
Assessment Plan for External Component			
Assessment Tool		COs Coverage	Weightage of Marks in External Component
Mid Term - I		CO1 (10 marks) + CO2 (5 marks)	15
Mid Term -II		CO2 (5 marks) + CO3 (10 Marks)	15
End Term		CO1+CO2+CO3+CO4+CO5 (4+4+4+14+14 Marks)	40
Total Marks			70

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
BTECH	IV	Design and Development of Cloud Services	25VA402	2	0	2	3
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
24	0	24	42		90		

Course Objective (s)		
The course aims:		
1. Understand cloud computing concepts, service models, deployment models, and cloud architecture principles.		
2. Design scalable, secure, and highly available cloud-based applications using modern architectural patterns.		
3. Apply cloud networking, storage, database, and security concepts for developing cloud services.		
4. Understand cloud migration strategies, cost optimization, governance, and sustainability principles		
5. Compare major cloud platforms (AWS, Microsoft Azure, Google Cloud Platform) and understand vendor-neutral cloud design practices.		
Syllabus		
Unit No.	Topics	CI Hours
1	Introduction to Cloud Architecture Cloud Architecture Principles, Cloud Reference Architecture, Cloud Design Patterns, Scalability, Elasticity, High Availability, Fault Tolerance, Cloud-native Applications, Monolithic and Microservices Architecture, Event-driven Architecture, RESTful Services, Best Practices for Cloud Application Design.	8
2	Containerization and Cloud Computing Technologies: Containerization Concepts, Docker Architecture, Docker Images and Containers, Container Lifecycle Management, Container Orchestration Concepts, Kubernetes Architecture (Introduction), Serverless Computing Concepts, Infrastructure as Code (IaC) Concepts. Lab: Exploring AWS Management Console and Cloud Architecture.	8
3	Advanced Cloud Networking and DevOps Virtual Networking Concepts, Software Defined Networking (SDN), Network Security, Load Balancing, Content Delivery Networks (CDN), API Management, DevOps Fundamentals, Continuous Integration (CI), Continuous Delivery (CD), Version Control Systems. Lab: Configure Security Group Inbound and Outbound Rules	8
4	Cloud Data Management and Security Cloud Database Services, Distributed Databases, Data Lakes, Cloud Data Analytics, Data Migration, Backup and Disaster Recovery, Identity and Access Management (IAM), Authentication and Authorization, Encryption, Monitoring, Logging and Auditing. Lab: Deploying a Static Website using Amazon S3.	8
5	Cloud Migration, Governance and Emerging Trends	8

	Cloud Migration Strategies, Cloud Cost Optimization, Resource Monitoring, Cloud Governance, Compliance, Multi-cloud and Hybrid Cloud Concepts, Edge Computing, Internet of Things (IoT) and Cloud Integration, Artificial Intelligence and Machine Learning on Cloud, Industry Case Studies. Lab: Deploying a Dynamic Website using Amazon EC2 and Monitoring Resources.	
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Prescribed Text Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Cloud Computing: Concepts, Technology & Architecture	Thomas Erl, Ricardo Puttini, Zaigham Mahmood	1st Edition	Pearson Education	2013
2	Cloud Computing: Principles and Paradigms	Rajkumar Buyya, James Broberg, Andrzej Goscinski	1st Edition	Wiley	2013
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	AWS for Solutions Architects	Saurabh Shrivastava, Neelanjali Srivastav	2nd Edition	Packt Publishing	2020
Web References					
1.	https://docs.aws.amazon.com/				
2.	https://aws.amazon.com/training/				
3.	https://learn.microsoft.com/en-gb/azure/				

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs):		
Upon the Completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Explain the fundamental concepts of cloud computing, cloud architecture, service models, and deployment models.	K2
CO2	Design cloud-based applications using cloud-native architecture and modern application development principles.	K3
CO3	Apply cloud networking, storage, and database concepts to develop scalable and efficient cloud solutions.	K3
CO4	Implement security, reliability, and disaster recovery principles to design secure and fault-tolerant cloud services.	K3
CO5	Evaluate cloud migration strategies, cost optimization techniques, and governance practices for efficient cloud adoption.	K5

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CO1	3	2	2	1	1	3	3	2	2	1	2			
CO2	1	3	2	1	2	1	3	1		3	1			

CO3	1	2	2	1		3		2	2	1	2			
CO4	1	1	2	1	2		3	1		3	1			
CO5	1	2	2	1		3	1		2	1	2			
Course	1.4	2	2	1	1.6	2.5	2.5	1.5	2	1.8	1.6			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern				
External Component			Internal Component	Total Marks
MT1	MT2	End Term	CIA	
15	15	40	30	100

Assessment Plan for CIA			
CO	Assessment Tool	Remarks	Weightage of Marks in CIA
CO1	Quiz, Debugging	50% + 50%	5
CO2	Quiz, Debugging	50% + 50%	5
CO3	In house competition participation	100%	5
CO4	Debugging, competition participation	30% + 70%	5
CO5	Debugging, competition participation	30% + 70%	5
	Attendance	100%	5
Total Marks			30
Assessment Plan for External Component			
Assessment Tool		COs Coverage	Weightage of Marks in External Component
Mid Term - I		CO1 (10 marks) + CO2 (5 marks)	15
Mid Term -II		CO2 (5 marks) + CO3 (10 Marks)	15
End Term		CO1+CO2+CO3+CO4+CO5 (4+4+4+14+14 Marks)	40
Total Marks			70

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	IV	Computer Organization and Architecture Lab	25CS451	0	0	2	1
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
0	0	24	06		30		

Course Objective (s)		
The course aims:		
1. To understand the behavior of logic gates, adders, decoders, multiplexers and flip flops.		
2. To measure the behavior of ALU and Processor from working modules along with the modules designed by the student as part of the experiment.		
3. To get familiarize with the hardware components of the computer system.		
Syllabus		
Exp. No.	List of Experiments	LI Hours
1	To Implement the 3–8-line DECODER	2
2	To simulate the 4x1 MULTIPLEXERS.	2
3	To analyse the Binary to Gray & Gray to Binary code conversions.	2
4	To design the Binary to BCD and BCD to Binary converter.	2
5	To implement HALF ADDER, FULL ADDER using basic logic gates.	2
6	To simulate the behaviour of Ripple Carry Adder.	2
7	To simulate the Carry Look Ahead Adder.	2
8	To verify the excitation tables of FLIP-FLOPS. (SR & D).	2
9	To verify the excitation tables of FLIP-FLOPS. (JK & T).	2
10	To implement HALF Subtractor, FULL Subtractor using basic logic gates.	2
Exp. No.	Value Added Experiments	LI Hours
11	To design a 1- bit ARITHMETIC LOGIC UNIT.	2
12	To simulate the magnitude comparator of 4 bit numbers.	2

Prescribed Textbooks					
S. No.	Book Title	Authors	Edition	Publisher	Year

1	Computer System Architecture	Morris Mano	3rd	Pearson	2017
2	Computer System Architecture, Revised 3/e Computer System Architecture, Designing for Performance	William Stallings	11th	Pearson	2022
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Fundamentals of Computer Organization and Architecture	Jyotsna Sengupta	1st	Deep & Deep Publications	2009
2	Computer Architecture and Organization	John P. Hayes	3rd	TMH	1998
Web References					
1. https://nptel.ac.in/courses/106106166					

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs):		
Upon the completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Demonstrate proficiency in digital simulation tools by implementing and verifying the truth table and behaviour of basic logic gates.	K3
CO2	Design and analyse combinational logic circuits such as decoders, multiplexers, and code converters (Binary to Gray/ BCD) to solve specific data routing and transformation requirements.	K5
CO3	Design and analyse sequential logic circuits such as flip-flops (SR, JK, T, and D)	K5
CO4	Implement and evaluate arithmetic circuits, including various adders (Half, Full, Ripple Carry, and Carry Look Ahead) & subtractors architectures (Half & Full).	K5
CO5	Design Complex Systems: Design and simulate a 1-bit Arithmetic Logic Unit (ALU).	K6

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	3	2	2	3			2		1	1	1		1
CO2	2	3	3	3	3			2		2	2	1		2
CO3	3	3	2	2	3			2		2	2	1		2

CO4	3	3	2	2	3			2		2	1	1		2
CO5	2	3	2	3	3			2		1	1	1		1
Course Avg.	2.60	3.00	2.20	2.40	3.00			2.00		1.60	1.40	1.00		1.60

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	100

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Quiz 1 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO2			
	Attendance		100%	5
PA2	CO3	Quiz 2 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO4			
	CO5			
	Attendance		100%	5
Total Marks				50
Assessment Plan for External Component				
Assessment Tool			COs Coverage	Weightage of Marks in External Component
End Term			All COs	50
Total Marks				50

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	IV	DBMS Lab	25CS452	0	0	2	2
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
0	0	24	06		30		

Course Objective (s)		
The course aims:		
1. To design relational databases using ER diagrams and SQL DDL commands with constraints.		
2. To perform database operations using SQL queries, joins, subqueries, grouping, views, and indexes.		
3. To develop a database application by applying DBMS concepts in a mini project.		
Syllabus		
Exp. No.	List of Experiments	LI Hours
1	Creating ER Diagram using Case Tools	2
2	Reduction of ER Diagram into Relational Model.	2
3	Execution of DDL Commands and Constraints	2
4	Execution of DML Commands	2
5	Execution of DQL Commands & Sorting Data	2
6	Implementation of Subqueries/Nested Query	2
7	Implementation of Joins using multiple table	2
8	Execution of Group by & having clause for Restriction data	2
9	Execution of Views and Indexes	4
10	Mini project (Design & Development of Data and Application)	4

Prescribed Textbooks					
S. No.	Book Title	Authors	Edition	Publisher	Year

1	Database Concepts	Korth, Silbertz, Sudarshan	7th	McGraw Hill	2019
2	Fundamentals of Database Systems	Elmasri, Navathe	7th	Addison Wesley	2015
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Database Management Systems	RAMAKRISHNAN	3rd	McGraw Hill	2002
2	An Introduction to Database Systems	Bipin C. Desai	Revised	Galgotia Publications	2010
Web References					
1. Data Base Management System – IIT Kharagpur (Prof. Partha Pratim Das, Prof. Samiran Chattopadhyay) https://onlinecourses.nptel.ac.in/noc26_cs39/preview					
2. Database Management System Part-1 https://infyspringboard.onwingspan.com/web/en/app/toc/lex_auth_01275806667282022456_shared/overview					

***Certification for the self-learning component should preferably be included in the assessment.**

Course Outcomes (COs):		
Upon the completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Understand and design database structures by creating ER diagrams and converting them into relational models using case tools.	K3
CO2	Apply DDL and DML commands to create, modify, and manipulate database objects with appropriate constraints.	K3
CO3	Analyze and execute DQL commands, subqueries, and joins for retrieving and managing data from multiple tables.	K4
CO4	Implement grouping operations, HAVING clause, views, and indexes for efficient data restriction and database optimization.	K3
CO5	Design and develop a mini database application project by integrating database concepts and SQL operations.	K6

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	3	2	2	3	2	-	2	3	-	2			
CO2	3	3	3	3	3	2	-	3	3	-	2			

CO3	3	3	3	3	3	2	-	3	2	-	2			
CO4	3	3	3	3	3	2	-	1	2	-	2			
CO5	3	3	3	3	3	2	-	1	2	-	2			
Course Avg.	3.00	3.00	2.80	2.80	3.00	2.00		2.00	2.40		2.00			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	100

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Quiz 1 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO2			
	Attendance		100%	5
PA2	CO3	Quiz 2 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO4			
	CO5			
	Attendance		100%	5
Total Marks				50
Assessment Plan for External Component				
Assessment Tool			COs Coverage	Weightage of Marks in External Component
End Term			All COs	50
Total Marks				50

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B.Tech.	IV	Design and Analysis of Algorithm Lab	25CS453	0	0	2	1
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL				
0	0	24	06*		30		

* *Participation in online coding contests on popular coding platforms.*

Course Objective (s)		
The course aims		
1.	The objective of this course is to provide hands-on experience in implementing and debugging algorithmic techniques such as Sliding Window, Prefix Sum, Two Pointers, Greedy, Backtracking, and Dynamic Programming.	
2.	The course aims to develop efficient problem-solving skills with optimized execution time and memory usage.	
3.	To develop the ability to evaluate algorithmic solutions by analyzing their correctness, time complexity, and space complexity, and comparing alternative approaches.	
Syllabus		
ExP. No.	List of Experiments	LI Hours
1	Given a word pat and a text txt, count the number of occurrences of anagrams of the word in the text. Also, given a string s and an integer k, find the length of the longest substring containing at most k distinct characters.	2
2	Given two strings s and t, find the smallest substring of s that contains all the characters of t (including duplicates). If no such substring exists, return an empty string (""). The answer is guaranteed to be unique.	2
3	Given an array arr[] containing 0s and 1s, count the number of subarrays having an equal number of 0s and 1s. Also, given two integer arrays L[] and R[], where L[i] and R[i] represent the start and end of a range, determine the integer that occurs in the maximum number of ranges. If multiple integers have the same maximum frequency, return the smallest one.	2
4	Given an integer array nums and an integer target, find all unique quadruplets [nums[a], nums[b], nums[c], nums[d]] such that the four indices are distinct and their sum equals the target. The output may be returned in any order.	2
5	Given start[] and finish[] arrays representing the start and finish times of activities, determine the maximum number of non-overlapping activities that can be performed by a single person. An activity starting at time x cannot be selected if another activity finishes at time x.	2
6	Given N points on a 2D plane, where the cost of connecting two points is their Manhattan distance, find the minimum total cost required to connect all points such that there is exactly one simple path between any two points. Also, given N cities and a list of flights, flights[i] = [from, to,	2

	price], determine the minimum cost to travel from src to dst using at most k stops. If no such route exists, return -1.	
7	Given an integer array nums that may contain duplicates, generate all unique subsets (the power set) without duplicate subsets. Also, given a string inStr, generate all possible palindrome partitions such that every substring in each partition is a palindrome.	2
8	Given an integer n, solve the N-Queens problem by placing n queens on an n × n chessboard such that no two queens attack each other. Return all distinct valid board configurations, where 'Q' represents a queen and '.' represents an empty space.	2
9	Given an array cost[], where cost[i] represents the cost of stepping on the ith stair, determine the minimum cost to reach the top when you can climb either one or two steps at a time, starting from step 0 or step 1. Also, given two strings s1 and s2, find the length of their Longest Common Subsequence (LCS). If no common subsequence exists, return 0.	2
10	Given an undirected connected graph with n nodes, determine the length of the shortest path that visits every node. The path may start from any node, revisit nodes multiple times, and reuse edges.	2
Exp. No.	Value Added Experiments	LI Hours
11	Develop a text analysis program that processes a given paragraph and performs the following operations: 1. Find the smallest window substring containing all characters of a given keyword. 2. Count the number of anagram occurrences of the keyword in the text. 3. Determine the length of the longest substring containing at most K distinct characters. Display the results of all three operations.	2
12	Design a route planning system for a transportation network represented as a graph. The program should: 1. Find the minimum cost to travel between two cities with at most K stops. 2. Determine the minimum cost required to connect all cities (or locations) using a Minimum Spanning Tree (MST). 3. Display the optimal route cost and the total network connection cost.	2

Prescribed Text Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Introduction to Algorithms	Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest and Clifford Stein	4 th	The MIT Press	2022
2	Guide to Competitive Programming: Learning and Improving Algorithms Through Contests	Antti Laaksonen	2 nd	Springer	2020

Reference Books

S. No.	Book Title	Authors	Edition	Publisher	Year
1	The Algorithm Design Manual	Steven S. Skiena	3rd	Springer	2020
2	Data Structures & Algorithms	Alfred V. Aho, John E. Hopcroft and Jeffrey D. Ullman	1st	Pearson Education	2002

Web References

- https://onlinecourses.nptel.ac.in/e-learning/preview/noc26_cs110

Course Outcomes (COs):

Upon the Completion of this course, the student will be able to:

CO No.	Statement	Knowledge Level, KL
CO1	Implement and debug sliding window algorithms to optimize execution time and memory usage.	K4
CO2	Implement and debug Prefix Sum and Two Pointer approach to optimize execution time and memory usage.	K4
CO3	Implement and debug Greedy algorithms to optimize execution time and memory usage.	K4
CO4	Implement and debug Backtracking approach to optimize execution time and memory usage.	K4
CO5	Implement and debug Dynamic Programming approach to optimize execution time and memory usage.	K4

KL- Bloom's Knowledge Level ($K_1, K_2, K_3, K_4, K_5, K_6$)

K_1 - Remember, K_2 - Understand, K_3 - Apply, K_4 - Analyze, K_5 - Evaluate, K_6 - Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	1	2	2	2				1		2			
CO2	3	1	2	2	2				1		2			
CO3	3	1	2	2	2				1		2			
CO4	3	1	2	2	2				1		2			
CO5	3	1	2	2	2				1		2			

Course	3.00	1.00	2.00	2.00	2.00				1.00		2.00			
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3 – High; 2 – Medium; 1 – Low

Evaluation Pattern			
Internal Component (CIA)		External Components	Total Marks
PA1	PA2		
25	25	50	100

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Record of programs written during labs + coding contest.	20% + 80%	20
	CO2			
	Attendance		100%	5
PA2	CO3	Record of programs written during labs + coding contest.	20% + 80%	20
	CO4			
	CO5			
	Attendance		100%	5
Total Marks				50
Assessment Plan for External Component				
Assessment Tool		COs Coverage	Weightage of Marks in External Component	
Coding Contest (30%) + Viva (20%)		All COs	50	
Total Marks			50	

Program	Semester	Course Name	Course Code	Periods			Credit
				L	T	P	
B. Tech	IV	FSD Workshop-II	25VA451	0	0	2	1
Teaching & Learning Scheme (in hours per semester)							
Classroom Instruction (CI)		Lab Instruction (LI)	Term Work (TW) and Self Learning (SL) (TW+ SL)		Total no. of Hours per Semester		
L	T	P	SL*				
0	0	24	06		30		

Course Objective (s)		
The course aims:		
1.	To introduce students to containerization technologies and orchestration tools for application deployment using Docker and Kubernetes.	
2.	To familiarize students with concepts of Generative AI, Large Language Models, and prompt engineering using platforms such as ChatGPT and Claude.	
3.	To develop understanding of DevOps practices including Continuous Integration and Continuous Deployment using tools like GitHub Actions and GitLab CI.	
4.	To enable students to design microservices-based applications and implement full-stack systems with containerized environments.	
5.	To equip students with skills to integrate AI capabilities and deploy applications on cloud platforms with monitoring and logging mechanisms.	
Syllabus		
Exp. No.	List of Experiments	LI Hours
1	Install Docker & Kubernetes (Minikube or K3s) Create a simple Dockerfile for a Node.js Build and run a Docker container locally Write YAML to create a basic Kubernetes Pod and Service	4
2	Research 3 use cases of Generative AI in real life Use OpenAI or Hugging Face to build a simple text generation tool Try a GenAI tool (like GitHub Copilot or ChatGPT) and reflect on your experience Submit: A mini blog + working code	4
3	Create a GitHub repository with a basic app (HTML+JS, Node.js) Setup GitHub Actions to run a build test on every push Dockerize the app and deploy it on Render/Heroku/Railway (free tier) Submit: GitHub repo + live app link + screenshot of workflow running	6
4	Submit team details & finalized project topic Project charter (goal, problem statement, features) Initial architecture diagram	4

	Set up project board (GitHub Projects) Submit detailed system design document Define all APIs (with request/response) Database schema design (ERD or table structure) Update project board with tasks assigned per member Submit backend API endpoints (at least 3)	
5	Implement DB connection Dockerize backend service Push image to DockerHub (optional) Build the frontend interface (basic working UI) Connect frontend to backend APIs Deploy locally using Kubernetes (1 service + 1 pod minimum) Submit: Screenshots + repo link	6
Exp. No.	Value Added Experiments	LI Hours
6	Add GenAI component (e.g., chatbot, content generator) Integrate OpenAI or Hugging Face API Test AI responses with sample inputs Setup GitHub Actions for testing & deployment Deploy app to staging server (Render/Heroku/VM) Submit: GitHub Actions YAML + deployment link	4
7	Full system deployed on cloud Add basic monitoring using Grafana or any logging tool Create & share a dashboard link or screenshot Submit: Cloud link + monitoring setup proof Perform usability & load testing (use tools like Postman, JMeter optional) Fix critical bugs Prepare: User manual, Installation guide, Final project video Final code submission on GitHub	2

Prescribed Textbooks					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Web Technologies – A Developer’s Perspective	N.P. Gopalan and J. Akilandeswari	3rd	Pearson	2025
2	Node.js, Express, MongoDB & JavaScript	Brad Dayley	11th	Pearson	2024
Reference Books					
S. No.	Book Title	Authors	Edition	Publisher	Year
1	Learning Web Design	Jennifer Robbins	5th	Mc Graw Hill	2012
Web References					
1. AWS Academy – https://aws.amazon.com/training/ 2. Microsoft Learn – https://learn.microsoft.com 3. Google Cloud Skills Boost – https://cloudskillsboost.google 4. Coursera – https://www.coursera.org 5. edX – https://www.edx.org					

**Certification for the self-learning component should preferably be included in the assessment.*

Course Outcomes (COs):		
Upon the completion of this course, the student will be able to:		
CO No.	Statement	Knowledge Level, KL
CO1	Understand the concepts of containerization, orchestration, and cloud deployment using tools such as Docker and Kubernetes	K2
CO2	Understand the fundamentals of Generative AI, Large Language Models, and prompt engineering using platforms like ChatGPT and Claude.	K2
CO3	Apply DevOps practices to implement Continuous Integration and Continuous Deployment pipelines using GitHub Actions or GitLab CI.	K3
CO4	Design and develop a containerized microservices-based application with integrated frontend and backend components.	K3
CO5	Build and deploy a full-stack application with integrated Generative AI features and monitoring mechanisms on cloud platforms.	K3

KL- Bloom's Knowledge Level (K₁, K₂, K₃, K₄, K₅, K₆)

K₁- Remember, K₂- Understand, K₃- Apply, K₄- Analyze, K₅- Evaluate, K₆- Create

CO-PO & PSO Mapping:

CO No.	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO1	PSO2	PSO 3
CO1	3	2			2						1			
CO2	2	2			1									
CO3	2	2	2		3			1	1		2			
CO4	2	3	3	2	3			1	2	1	2			
CO5	2	3	3	2	3	1		1	2	2	3			
Course Avg.	2.20	2.40	1.60	0.80	2.40	0.20		0.60	1.00	0.60	1.60			

3 – High; 2 – Medium; 1 – Low

Evaluation Pattern			
Internal Component (CIA)		External Component	Total Marks
PA1	PA2		
20 +5 (Attendance)	20 +5 (Attendance)	50	
			100

Assessment Plan for CIA				
	CO	Assessment Tool	Remarks	Weightage of Marks in CIA
PA1	CO1	Quiz 1 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO2			
	Attendance		100%	5
	CO3	Quiz 2 + Conduction of Experiment + Records/ Viva	25% + 25% + 50%	20
	CO4			

PA2	CO5			
	Attendance		100%	5
Total Marks				50
Assessment Plan for External Component				
Assessment Tool			COs Coverage	Weightage of Marks in External Component
End Term			All COs	50
Total Marks				50